

# Dongfang Zhao, Ph.D.

Nationality: United States of America

## Education

|                                           |                           |      |
|-------------------------------------------|---------------------------|------|
| University of Washington, Seattle         | Computer Science, PostDoc | 2017 |
| Illinois Institute of Technology, Chicago | Computer Science, PhD     | 2015 |
| Emory University, Atlanta, GA             | Computer Science, MS      | 2008 |
| Katholieke Universiteit Leuven, Belgium   | Statistics, Diploma       | 2005 |

## Employment

|           |                                                                                                                                                                                                                                                                      |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023–now  | University of Washington (UW) <ul style="list-style-type: none"><li>◦ Tenure-Track Assistant Professor, UW Tacoma School of Engineering &amp; Technology</li><li>◦ Adjunct Assistant Professor, Paul G. Allen School of Computer Science &amp; Engineering</li></ul> |
| 2018–2023 | Assistant Professor, University of Nevada, Reno                                                                                                                                                                                                                      |
| 2015      | Research Intern, IBM Almaden Research Center, San Jose, CA                                                                                                                                                                                                           |
| 2014      | Research Intern, Argonne National Laboratory, Chicago, IL                                                                                                                                                                                                            |
| 2013      | Research Intern, Pacific Northwest National Laboratory, Seattle, WA                                                                                                                                                                                                  |
| 2009–2011 | Software Developer (full-time), Epic Systems Corporation, Madison, WI                                                                                                                                                                                                |

## Awards

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|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 | Federal Government Research Grant, NSF 2112345, \$255,916<br>“A DLT Machine Learning Platform for Blockchain Warehousing”                                                |
| 2020 | Federal Government Research Grant, DOE SC0020455, \$200,000<br>“HPChain: Energy-Efficient Scalable Blockchains for HPC Systems”                                          |
| 2019 | Best Paper Award, International Conference on Cloud Computing<br>“BlockLite: Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud”                 |
| 2018 | Best Student Paper Award, IEEE International Conference on Cloud Computing<br>“Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory” |

## Recent Projects

|                     |                                                                                                                                                                                                                                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>[BadRSSD]</b>    | Information-theoretical backdoor attacks on diffusion models utilizing residual score subspace dispersive loss: <a href="https://github.com/hpdic/BadRSSD">https://github.com/hpdic/BadRSSD</a> (private for now)<br><i>Tags:</i> Generative AI security, Latent subspace learning, Fine-tuning integrity, PyTorch |
| <b>[ElasticIVF]</b> | Elastic in-memory vector indexing and search for streaming data analysis on heterogeneous distributed platforms: <a href="https://github.com/hpdic/ElasticIVF">https://github.com/hpdic/ElasticIVF</a><br><i>Tags:</i> Distributed GPU computing, Message passing interface (MPI), Faiss, C++, Python              |
| <b>[AdaDisk]</b>    | Decentralized agentic systems for adaptive ingestion-query scheduling of disk-resident vector indices in LLM serving: <a href="https://github.com/hpdic/AdaDisk">https://github.com/hpdic/AdaDisk</a><br><i>Tags:</i> Large language models, AI agents, Distributed systems, DiskANN, C++, Python                  |
| <b>[Hermes]</b>     | MySQL plug-in for high-performance homomorphic encryption of vector databases in privacy-preserving RAG: <a href="https://github.com/hpdic/Hermes">https://github.com/hpdic/Hermes</a><br><i>Tags:</i> Retrieval-augmented generation, Fully homomorphic encryption, OpenFHE, C++                                  |

## Selected Publications (Dongfang Zhao was/is the \*student's academic advisor)

- [1] \*Baigang Chen, **Dongfang Zhao**. Reliable Non-Leveled Homomorphic Encryption for Web Services. The ACM Web Conference (formerly International World Wide Web Conference, WWW), 2026.
- [2] \*Chengyu Gong, \*Gefei Shen, Luanzheng Guo, Nathan Tallent, **Dongfang Zhao**. Order-Preserving Dimension Reduction for Multimodal Semantic Embedding. The 40th Annual AAAI Conference on Artificial Intelligence (AAAI), 2026.
- [3] \*Jiuzhou Fu, Luanzheng Guo, Nathan Tallent, **Dongfang Zhao**. ProHD: Projection-Based Hausdorff Distance Approximation. IEEE International Conference on Data Mining (ICDM), 2025.
- [4] \*Jiayao Wang, . . . , **Dongfang Zhao**. IPBA: Imperceptible Perturbation Backdoor Attack in Federated Self-Supervised Learning. European Conference on Artificial Intelligence (ECAI), 2025.
- [5] \*Olamide Timothy Tawose, Jun Dai, Lei Yang, **Dongfang Zhao**. Toward Efficient Homomorphic Encryption for Outsourced Databases through Parallel Caching. Proceedings of the ACM on Management of Data (SIGMOD), 2023.
- [6] \*Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. SciChain: Blockchain-enabled Lightweight and Efficient Data Provenance for Reproducible Scientific Computing. 37th IEEE International Conference on Data Engineering (ICDE), 2021.
- [7] \*Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. BAASH: lightweight, efficient, and reliable blockchain-as-a-service for HPC systems. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2021.
- [8] \*Xinying Wang, \*Olamide Timothy Tawose, Feng Yan, **Dongfang Zhao**. HDK: Toward High-Performance Deep-Learning-Based Kirchhoff Analysis. The Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI), 2020.
- [9] Heyang Qin, Syed Zawad, Yanqi Zhou, Lei Yang, **Dongfang Zhao**, Feng Yan. Swift machine learning model serving scheduling: a region based reinforcement learning approach. The International Conf. for High Performance Computing, Networking, Storage and Analysis (SC), 2019.
- [10] Parmita Mehta, Sven Dorkenwald, **Dongfang Zhao**, Tomer Kaftan, Alvin Cheung, Magdalena Balazinska, Ariel Rokem, Andrew J. Connolly, Jacob VanderPlas, Yusra AlSayyad. Comparative Evaluation of Big-Data Systems on Scientific Image Analytics Workloads. Proceedings of Very Large Data Bases (VLDB), 2017.
- [11] **Dongfang Zhao**, Kan Qiao, Jian Yin, Ioan Raicu. Dynamic Virtual Chunks: On Supporting Efficient Accesses to Compressed Scientific Data. IEEE Transactions on Services Computing (TSC). 9(1): 96-109 (2016).
- [12] **Dongfang Zhao**, Ning Liu, Dries Kimpe, Robert B. Ross, Xian-He Sun, Ioan Raicu. Towards Exploring Data-Intensive Scientific Applications at Extreme Scales through Systems and Simulations. IEEE Transactions on Parallel and Distributed Systems (TPDS). 27(6): 1824-1837 (2016).
- [13] **Dongfang Zhao**, Li Yang. Incremental Isometric Embedding of High-Dimensional Data Using Connected Neighborhood Graphs. IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI). 31(1): 86-98 (2009).